

Ajay Kumar Reddy Nelavetla

Silicon Valley, CA | +1 857-576-1177 | ajaykumarreddynelavetla@gmail.com | [LinkedIn](#) | [GitHub](#)

AI/ML Engineer with 5+ years building production-grade generative AI systems and multi-agent architectures at scale. Specialized in LLM applications, RAG implementations, agentic AI frameworks, and end-to-end deployment of intelligent systems. Proven track record shipping GenAI products serving 10,000+ users with strong focus on reliability and real-world impact.

PROFESSIONAL EXPERIENCE

AI/ML Engineer | FolderWave, Boston, MA

Jan 2024 – Present

- Engineered production real-time data pipeline with Apache Kafka (ETL) and Apache Spark processing 1000+ events/second with sub-100ms latency, extracting 50+ behavioral signals using feature engineering and dimensionality reduction for dynamic content personalization.
- Architected data infrastructure with Pinecone vector database achieving 95% retrieval accuracy, supporting ML-powered recommendation engine processing 500K+ daily user interactions with 30% engagement increase.
- Built RESTful API architecture with 400+ OpenAPI-compliant endpoints optimized for sub-100ms response times and GraphQL implementation, deployed on AWS (Lambda, ECS, SageMaker, Kinesis, RDS) with Kubernetes orchestration.
- Implemented enterprise-grade multi-tenant SaaS data architecture with OAuth2 authentication, RBAC, and strict data isolation, deployed automated CI/CD pipelines (GitHub Actions), CloudWatch monitoring, auto-scaling policies, achieving 99.9% uptime SLA.
- Optimized database performance through advanced partitioning strategies, composite indexing, query execution plan optimization, and statistics maintenance, deployed on AWS (RDS, S3) with automated failover supporting petabyte-scale data

Full Stack Developer | IdeaBytes, Hyderabad, India

Jan 2019 – Jan 2022

- Designed enterprise-grade data architecture with 400+ REST API endpoints featuring OAuth2 authentication, rate limiting, JWT tokens, and async processing to handle 10,000+ concurrent requests with comprehensive OpenAPI documentation.
- Created automated ETL workflows with Apache Spark batch processing and Hadoop integration for large-scale data transformation.
- Optimized database performance using advanced partitioning, composite indexing, and query optimization, deployed on AWS (EC2, S3, RDS) with ELK Stack logging, CloudWatch alerting, and 99.9% availability SLA.

PROFESSIONAL PROJECTS

Multi-Platform Social Media Automation SaaS | [Link](#)

Aug 2025 – Present

- Architected multi-tenant SaaS data infrastructure with OAuth2 integrations with 15+ social media platforms including Instagram, TikTok implementing RBAC for multi-level access control and intelligent data scheduling processing 500+ posts daily achieving 99.5% uptime.
- Built real-time data pipelines with Node.js/TypeScript microservices and serverless architecture, integrating vector databases (Pinecone, Weaviate) for semantic search and AI-powered content matching serving 100+ active users.
- Deployed on Railway with automated CI/CD pipelines, CloudWatch monitoring and horizontal auto-scaling supporting 100+ users

AI-Powered Job Application Automation | [GitHub](#)

Jan 2025 – July 2025

- Developed full-stack data extraction pipeline processing 100+ application types with 92% accuracy using web scraping (Selenium, BeautifulSoup), HTML parsing, DOM analysis, and adaptive extraction algorithms.
- Implemented vector database integration (Pinecone, Weaviate) for semantic resume-to-job matching achieving 85% relevance scoring with automated data validation and quality checks.
- Created comprehensive automation pipeline processing 100+ applications daily with intelligent filtering, data cleansing, and error recovery mechanisms.

AI Voice Agent

Sep 2024 – Dec 2024

- Built production real-time streaming data system with WebSocket communication achieving sub-100ms end-to-end latency, implementing optimized frame buffering (20ms intervals), audio codec handling (opus, PCM, AAC), and concurrent connection management supporting 50+ simultaneous sessions.
- Engineered low-latency data pipeline with parallel processing, audio optimization, and adaptive bitrate streaming, reducing response time by 60% through efficient data handling and caching strategies.

PROFESSIONAL SKILLS

- LLM & Generative AI:** GPT-4, Claude Sonnet 4, Llama-3, OpenAI API, Anthropic API | Prompt Engineering | Fine-tuning, LoRA, RLHF | LangChain, LlamaIndex | Function Calling | Token Optimization
- Data Engineering & Pipelines:** Apache Kafka, Apache Spark, Airflow, Prefect, Debezium, AWS Kinesis | ETL/ELT Workflows | Stream Processing | Real-time Data Pipelines | Event-Driven Architecture | Batch Processing
- Databases & Data Warehouses:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Cassandra | Snowflake, BigQuery | Query Optimization | Indexing & Partitioning | Data Modeling (OLTP/OLAP) | Dimensional Modeling
- Vector Databases & AI/ML Data Infrastructure:** Pinecone, Weaviate, Chroma, FAISS | RAG Systems | Semantic Search | Feature Engineering | Feature Stores | Model Serving | Data Versioning
- Cloud & Infrastructure:** AWS (RDS/Aurora, S3, Kinesis, Lambda, CloudWatch, SageMaker, ECS) | Azure (ML Studio, OpenAI Service) | GCP (Vertex AI) | Docker, Kubernetes | Terraform, CloudFormation | IAC | Auto-scaling | Cost Optimization
- Programming & Development:** Python (Expert), SQL, JavaScript/TypeScript, Java, C/C++, Go | REST APIs (400+ endpoints), GraphQL, WebSockets | FastAPI, Flask, Spring Boot, Node.js, Django | Microservices Architecture
- Data Quality & Observability:** Data Validation Frameworks | Monitoring & Alerting | CloudWatch, Prometheus, Grafana, ELK Stack, Datadog | Performance Tuning | Data Governance | Observability | Logging
- MLOps & AI Systems:** MLflow, Kubeflow, Weights & Biases | Model Monitoring | CI/CD Pipelines (GitHub Actions, Jenkins, GitLab CI) | Feature Store | Automated Retraining | Drift Detection
- Enterprise SaaS Architecture:** Multi-tenant Systems | Data Isolation | OAuth2, RBAC | API-First Architecture | Security & Compliance (SOC 2, GDPR, KYC/AML) | Identity & Access Management (IAM)

EDUCATION

Master of Science in Software Engineering | Northeastern University, Boston, MA

Jan 2022 – Dec 2023

Bachelor of Technology in Electronics and Communication Engineering | JBIT, Hyderabad, India

Sep 2016 – Sep 2020